Enterprise BPMN 2.0 Process Modelling & Defence

Comparative Process Modelling & Exception Architecture for Questions 1 & 2

• Q1: Student Project Approval & Allocation System | • Q2: Loan Origination & Approval System

CANDIDATE & ACADEMIC INFORMATION

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Course / Subject: Workflow Automation

Academic Institution: SRMIST

SCOPE & ARCHITECTURAL HIGHLIGHTS

Specification Standard: OMG BPMN 2.0 XML & Visual Diagram Interchange

Question 1 Architecture: 5 Swimlanes | 11 Failure Paths | DMN Guide Allocation

Question 2 Architecture: 7 Lanes (1 External Pool) | 12 Failure Paths | Parallel Concurrency

Engine Validation: Tested on Camunda Modeler 5.49 & bpmn-js Engine

PROJECT CONTEXT

Problem Statements & Engineering Objectives

QUESTION 1: STUDENT PROJECT APPROVAL SYSTEM

• **The Academic Challenge:** University departments manage hundreds of final-year capstone proposals. Manual paper flows lead to lost forms, plagiarism oversights, revision deadlocks, and guide workload imbalances.

• **Key Participants (5 Lanes):** Student/Team, Project Coordinator, Review Committee (incl. HoD), Faculty Guide, and Project Management System.

• **Core Objectives:** • Automate completeness & team rule validation.

• Implement similarity/duplicate topic check gate.

• Manage iterative committee revision cycles (SLA timers).

• Rule-based faculty guide workload allocation.

• **Why BPMN 2.0:** BPMN swimlanes eliminate ambiguity between student submissions, coordinator screening, and committee academic evaluation.

QUESTION 2: LOAN ORIGINATION & APPROVAL SYSTEM

• **The Banking Challenge:** Commercial lending workflows suffer from long processing times, customer abandonment, regulatory KYC/AML violations, external API downtime, and core banking transaction failures.

• **Key Participants (7 Lanes/Pools):** Applicant, Loan Officer, Underwriter, Compliance Team, Core Banking, Operations, and External Credit Bureau Pool.

• **Core Objectives:** • Concurrently execute KYC/AML, Credit Bureau, and DTI checks.

• Enable underwriter manual review & collateral re-valuation.

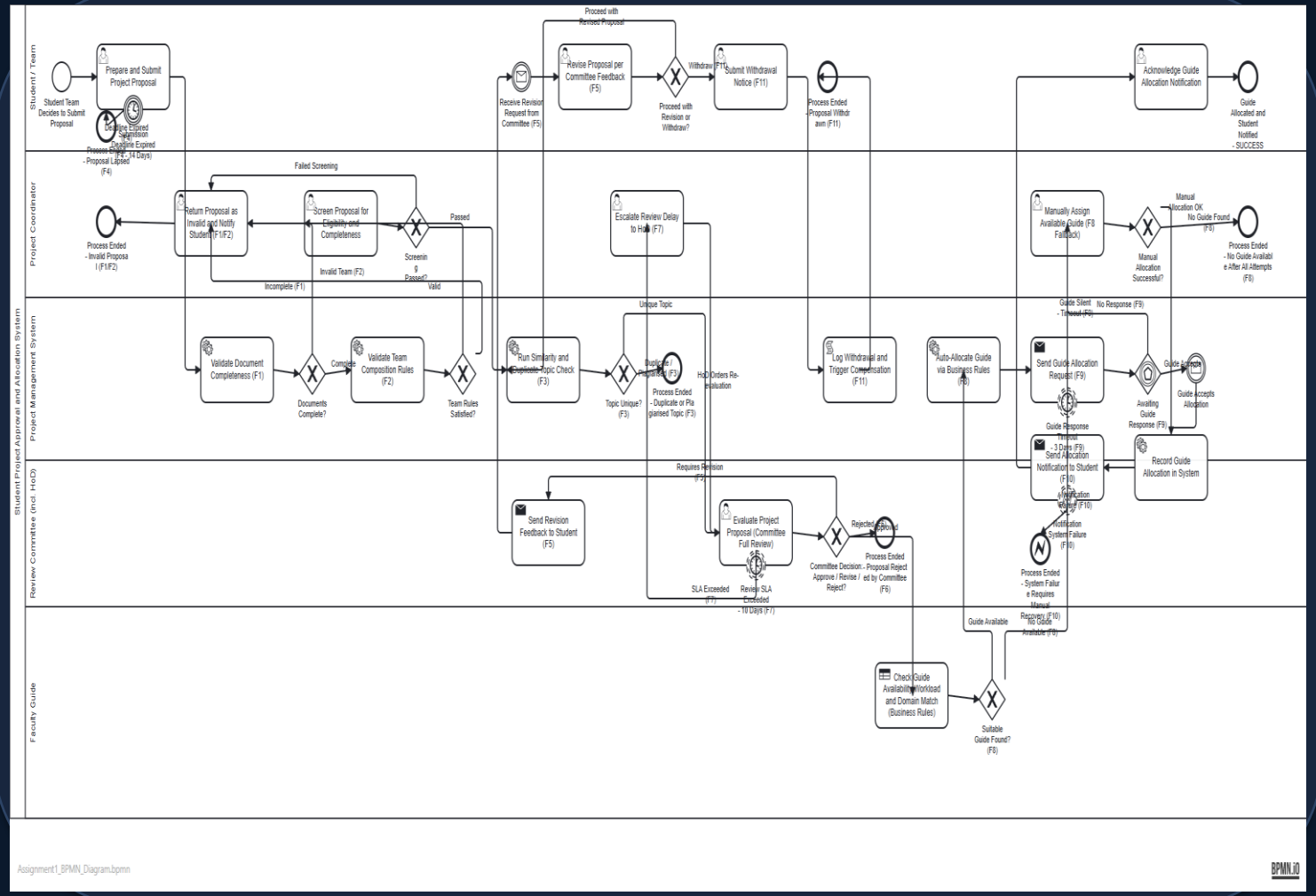
• Enforce 7-day offer acceptance expiry and 48-hour e-sign SLAs.

• Contain technical errors during NEFT/RTGS disbursement.

• **Why BPMN 2.0:** BPMN 2.0 provides formal pool separation for external partners and event boundaries for transaction safety.

SYSTEM ARCHITECTURE — Q1

Question 1: Student Project Approval Architecture



Q1 ARCHITECTURAL SUMMARY

✓ **Swimlanes (5):** Student/Team, Project Coordinator, Project Management System, Review Committee, Faculty Guide.

✓ **Process Boundaries:** Starts when student team submits proposal. Ends when guide is allocated & confirmed, or proposal rejected/lapsed.

✓ **Key Decision Gates:** 1. Document Completeness (System)

2. Team Rule Verification (System)

3. Similarity / Plagiarism Gate (System)

4. Committee Decision (Approve / Revise / Reject)

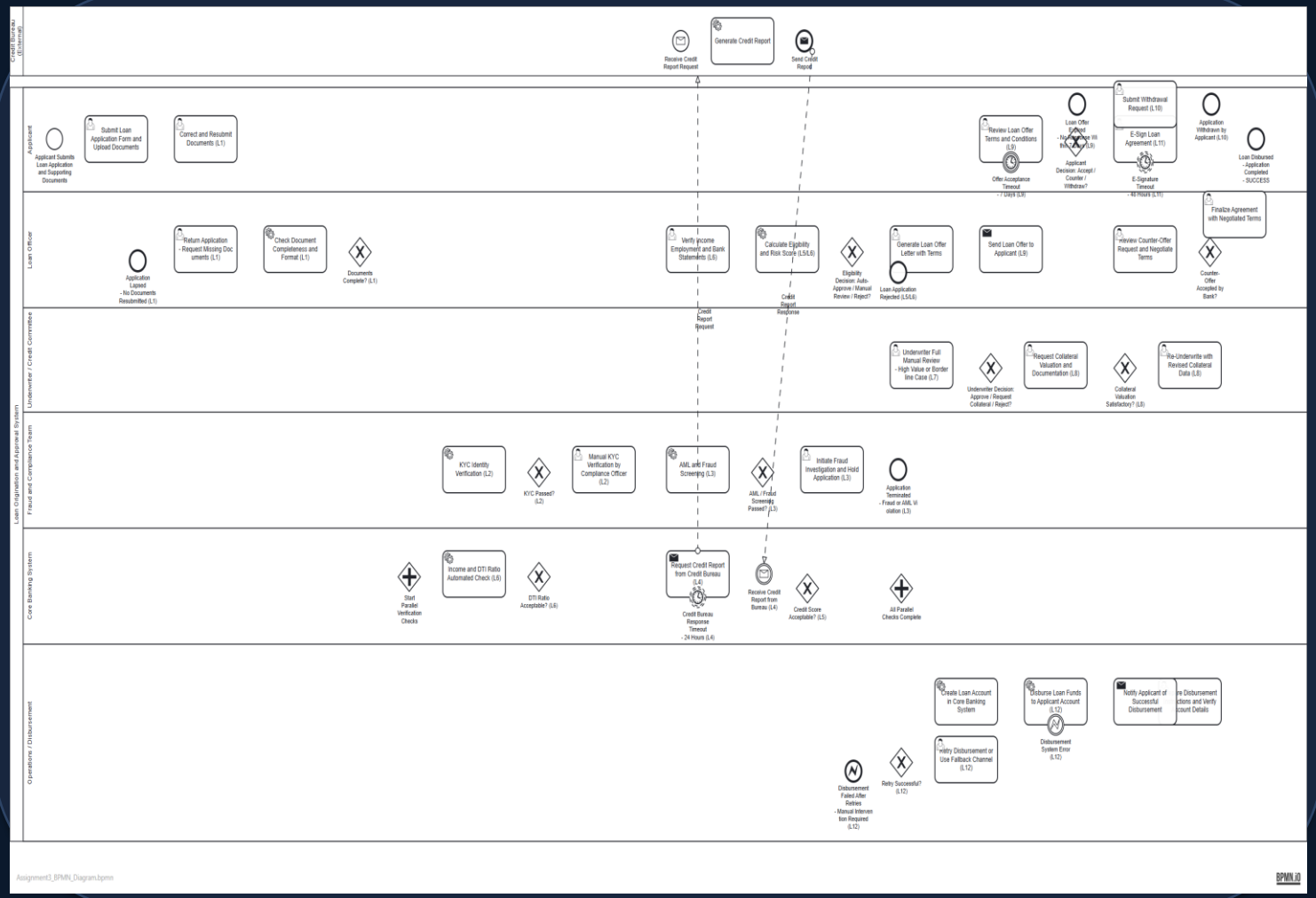
5. Guide Availability Match (DMN)

6. Guide Acceptance (Message / Timeout).

✓ **Failure End States (6):** Invalid Proposal, Duplicate Topic, Proposal Lapsed, Proposal Rejected, No Guide Available, System Failure.

SYSTEM ARCHITECTURE — Q2

Question 2: Loan Origination & Approval Architecture



Q2 ARCHITECTURAL SUMMARY

✓ **Participants & Pools:** 7 Total: 6 internal bank lanes (Applicant, Loan Officer, Underwriter, Compliance, Core Banking, Operations) + 1 External Pool (Credit Bureau).

✓ **Process Boundaries:** Starts when applicant submits loan form & docs. Ends when loan is disbursed, or rejected/withdrawn/expired/terminated.

✓ **Key Concurrency Feature:** Parallel Gateway forks 3 concurrent streams: KYC/AML check, Credit Bureau request, and DTI calculation.

✓ **External Message Exchange:** Message Flows connect Bank Core Banking lane to external Credit Bureau Pool for credit reports.

✓ **Failure End States (6):** Rejected, Withdrawn, Application Lapsed, Offer Expired, Fraud Terminated, Disbursement Failed.

BPMN CONSTRUCTS

Task Types Used Across Both BPMN Processes

👤 USER TASK

Def: Human interaction supported by UI

Q1 Examples:
- Prepare & Submit Proposal (Student)
- Screen Proposal (Coordinator)
- Evaluate Proposal (Committee)
- Manual Guide Allocation (Coordinator)

Q2 Examples:
- Submit Application (Applicant)
- Verify Income & Employment (Officer)
- Underwriter Full Review (Underwriter)
- E-Sign Loan Agreement (Applicant)

Why Chosen? Selected whenever task completion requires human judgement, evaluation, negotiation, or manual form inputs. Cannot be executed by an automated script.

⚙️ SERVICE TASK

Def: Automated system execution

Q1 Examples:
- Validate Document Completeness
- Validate Team Composition Rules
- Similarity & Duplicate Topic Check
- Auto-Allocate Guide

Q2 Examples:
- KYC Identity Verification API
- AML Watchlist Screening API
- Automated DTI Calculation
- Disburse Loan Funds (Core Banking)

Why Chosen? Selected for deterministic computational logic, API integrations, and database operations without human intervention. Ensures instant SLA execution.

✉️ / 📩 SEND TASK & CATCH

Def: Outbound message dispatch & asynchronous catch

Q1 Examples:
- Send Task: Guide Allocation Request
- Send Task: Revision Feedback
- Message Catch: Receive Guide Acceptance

Q2 Examples:
- Send Task: Request Credit Report
- Message Catch: Receive Credit Report
- Send Task: Send Loan Offer Letter

Why Chosen? Selected to model active dispatch and asynchronous waiting states across organizational or participant boundaries without thread-blocking.

📊 / 📜 BUSINESS RULE & SCRIPT

Def: Decision tables and procedural utilities

Q1 Business Rule Task:
- Check Guide Availability, Workload & Domain Match (Evaluates DMN decision table)

Q1 Script Task:
- Log Withdrawal & Release Guide Pre-Assignment (Internal compensation cleanup)

Q2 Service Scoring:
- Calculate Eligibility Score (Engine API)

Why Chosen? Business Rule Tasks decouple dynamic business policy (faculty workload limits) from process code. Script tasks perform rapid local memory cleanups.

CONTROL FLOW

Gateways & Concurrency Control (Q1 vs. Q2)

EXCLUSIVE GATEWAY (XOR ✕)

Decision Routing (Both Q1 & Q2)

Semantics: Evaluates condition expressions and activates EXACTLY ONE outgoing branch. On merge, passes any arriving token immediately.

- Q1: 'Proposal Valid?' (Valid vs Incomplete)
- Q1: 'Committee Decision' (Approve / Revise / Reject)
- Q1: 'Guide Available?' (Found vs Manual Allocation)
- Q2: 'KYC Passed?' (Pass vs Manual Review vs Terminate)
- Q2: 'Eligibility Decision' (Auto-Approve / Underwriter / Reject)
- Q2: 'Underwriter Decision' (Approve / Collateral / Reject)
- Q2: 'Offer Decision' (Accept / Counter / Withdraw)

Why Used? Loan decisions and proposal verdicts are mutually exclusive; an application cannot be simultaneously approved and rejected. XOR ensures deterministic execution.

PARALLEL GATEWAY (AND +)

Concurrency Synchronization (Q2)

Semantics: Forks one incoming path into multiple concurrent branches without conditions. The join waits for ALL incoming paths before proceeding.

- Q2 Fork: 'Start Parallel Verification Checks'
  - Branch 1: KYC Identity & AML Screening
  - Branch 2: Credit Bureau Score Retrieval
  - Branch 3: Debt-to-Income (DTI) Calculation
- Q2 Join: 'All Parallel Checks Complete'
  - Synchronizes all 3 branches before income verification.

Why Used? Running checks serially creates additive cycle time. Parallel execution cuts processing latency while the synchronized join prevents token race conditions.

EVENT-BASED GATEWAY (⬡)

Competing Event Selection (Q1)

Semantics: Routes based on whichever external event occurs first, rather than evaluated data expressions.

- Q1: 'Wait for Guide Acceptance or Timeout'
  - Branch A: Intermediate Message Catch ('Guide Accepts Request')
  - Branch B: Intermediate Timer Event ('3-Day Guide Response Timeout')
  - Result: If guide accepts, proceeds to record allocation; if 3 days elapse, diverts to Coordinator Manual Allocation.

Why Used? Models real-world asynchronous competition: the process cannot know in advance whether the faculty member will accept or ignore the request.

EVENT ARCHITECTURE

Intermediate & Boundary Events Deep-Dive

CRITICAL DISTINCTION: INLINE INTERMEDIATE EVENTS VS. TASK BOUNDARY EVENTS

- Inline Intermediate Events sit in the sequence flow and pause token progress until an external trigger arrives.
- Boundary Events attach to task perimeters; they monitor active execution and catch external timeouts, faults, or cancellations.

INLINE MESSAGE CATCH

Symbol: ✉ (Catch)
Type: Sequence Flow Intermediate Event

Q1: 'Receive Revision Request'
Student waits for committee feedback.
Q1: 'Receive Guide Acceptance'
System waits for guide response.
Q2: 'Receive Credit Report'
Core Banking waits for Credit Bureau report.

Why Needed? Models non-polling asynchronous waiting. Decouples collaborating parties.

NON-INTERRUPTING TIMER

Symbol: ◌ (Dashed Ring)
Type: Boundary Event on Task

Q1: '10-Day Review SLA Timer'
Fires HoD escalation reminder on committee.
Q2: '24-Hour Credit Bureau Timer'
Fires automated retry request.
Q2: '48-Hour E-Sign SLA Timer'
Triggers wet-signature alert to operations.

Why Needed? Fires parallel escalations without aborting or cancelling the primary active user task.

INTERRUPTING TIMER

Symbol: ◌ (Solid Ring)
Type: Boundary Event on Task

Q1: '14-Day Submission Deadline'
Cancels proposal prep → EndEvent: Lapsed.
Q1: '7-Day Revision Deadline'
Limits student revision window.
Q2: '7-Day Offer Validity Timer'
Cancels offer review → EndEvent: Offer Expired.

Why Needed? Enforces hard deadlines. Instantly cancels the underlying task and diverts flow to expiry end.

INTERRUPTING ERROR BOUNDARY

Symbol: ⚡ (Solid Ring)
Type: Boundary Event on Task

Q1: 'Notification Delivery Failure'
Catches portal/SMS outage → IT recovery.
Q2: 'Core Banking Disbursement Fault'
Catches NEFT/RTGS transaction failure on Disburse Funds task → Operations retry.

Why Needed? Traps technical and transaction exceptions, preventing untracked financial or system states.

EXECUTION LIFECYCLE

Happy Path & Business-Critical Flows (Q1 vs. Q2)

QUESTION 1 HAPPY PATH (ACADEMIC ALLOCATION)

1. Start Event: Student prepares and submits proposal form.
2. Service Task: System validates document completeness.
3. Service Task: System verifies team composition rules.
4. User Task: Coordinator screens proposal for eligibility.
5. Service Task: System executes similarity / duplicate check.
6. User Task: Review Committee evaluates and APPROVES proposal.
7. Business Rule Task: DMN checks guide workload and domain.
8. Service Task: System executes automated guide allocation.
9. Send Task: System dispatches allocation request to guide.
10. Event Catch: Guide accepts request within 3-day SLA.
11. Service Task: System formally records allocation in DB.
12. Send Task: System dispatches confirmation to student team.
13. User Task: Student acknowledges guide assignment.
14. SUCCESS END: Project Allocated & Guide Confirmed.

QUESTION 2 HAPPY PATH & TERMINOLOGY DEFENCE

1. Start Event: Applicant submits loan application & documents.
2. Service Task: Completeness check validates document schema.
3. Parallel Split: Triggers KYC/AML, Credit Bureau & DTI checks.
4. Parallel Join: Synchronizes all 3 successful background checks.
5. User Task: Loan Officer verifies income & employment.
6. Service Task: Scoring engine calculates Auto-Approval score.
7. Send Task: Officer generates & dispatches loan offer letter.
8. User Task: Applicant reviews & ACCEPTS offer within 7 days.
9. User Task: Applicant e-signs digital loan agreement.
10. Service Task & End: Core Banking disburses funds via NEFT/RTGS.

--- ACADEMIC CLARIFICATION ON CRITICAL PATH ---

- Happy Path = Ideal sequence with zero rework or failures.
- Critical Path Method (CPM) = Longest duration path with zero slack in operations research. It requires empirical task durations (PERT distributions).
- Academic Stance: We do NOT invent artificial task durations. We model the Happy Path as the 'Business-Critical Flow'.

FAILURE REGISTER — Q1

Failure Path Matrix: Question 1 (Student Project)

QUESTION 1 FAILURE PATH REGISTER (F1 – F6 HIGHLIGHTS)

[F1] Document Incompleteness: Trigger: Missing sections, absent signatures, wrong document format. | Detection: Service Task validation schema fails → Exclusive Gateway routes to Coordinator. | **Outcome: Correction Loop OR 'EndEvent: Invalid Proposal Returned'**

[F2] Team Rule Violation: Trigger: Team size below minimum, ineligible member, mixed-department rules. | Detection: Service Task validates rules → Exclusive Gateway branches to invalidation. | **Outcome: Terminal 'EndEvent: Invalid Proposal Returned'**

[F3] Duplicate / Plagiarised Topic: Trigger: Similarity engine detects ≥ 70% overlap with existing or past projects. | Detection: Service Task similarity API flags score → Exclusive Gateway routes to rejection. | **Outcome: Terminal 'EndEvent: Duplicate / Plagiarised Topic'**

[F4] Proposal Submission Lapsed: Trigger: Student team fails to submit proposal within the 14-day departmental window. | Detection: Interrupting Timer Boundary Event (14 days) fires on submission task. | **Outcome: Terminal 'EndEvent: Proposal Lapsed'**

[F5] Committee Revision Request: Trigger: Committee finds proposal technically weak or overly broad, but salvageable. | Detection: Exclusive Gateway on committee decision selects 'Requires Revision' branch. | **Outcome: Revision Loop back to Similarity Check OR Withdrawal**

[F6] Committee Rejection: Trigger: Proposal judged academically infeasible, out of scope, or non-viable. | Detection: Exclusive Gateway on committee decision selects 'Rejected' branch. | **Outcome: Terminal 'EndEvent: Proposal Rejected'**

FAILURE REGISTER — Q2

Failure Path Matrix: Question 2 (Loan Origination)

QUESTION 2 FAILURE PATH REGISTER (L1 – L6 HIGHLIGHTS)

[L1] Document Completeness: Trigger: Missing income proof, illegible PAN/Aadhaar scans, invalid file format. | Detection: Service Task completeness validation schema fails → Exclusive Gateway routes. | **Outcome: Correction Loop OR 'EndEvent: Application Lapsed'**

[L2] KYC Identity Mismatch: Trigger: Name misspelling, date of birth mismatch against government database. | Detection: KYC Service Task returns API error code → Exclusive Gateway branches. | **Outcome: Manual Clearance OR 'EndEvent: Fraud Terminated'**

[L3] AML Watchlist Flag: Trigger: Applicant matches PEP lists, sanction registries, or financial crime DB. | Detection: AML Service Task flags risk score ≥ threshold → Exclusive Gateway routes to probe. | **Outcome: Terminal 'EndEvent: Application Terminated – Fraud/AML'**

[L4] Credit Bureau Timeout: Trigger: Credit bureau service unresponsive, network partition, timeout > 24 hours. | Detection: Non-interrupting Timer Boundary Event (24h) fires on outbound request task. | **Outcome: Resilient Catch on retry OR manual credit assessment fallback**

[L5] Credit Score Rejection: Trigger: Report returns credit bureau score below minimum underwriting floor (< 600). | Detection: Credit Bureau gateway evaluates score; feeds failed attribute into parallel join. | **Outcome: Terminal 'EndEvent: Loan Rejected – Low Credit Score'**

[L6] Excessive DTI Ratio: Trigger: Applicant liabilities result in debt-to-income ratio exceeding 50% limit. | Detection: Automated DTI Service Task computes ratio > threshold → Exclusive Gateway routes. | **Outcome: Terminal 'EndEvent: Loan Rejected – Excessive DTI'**

BRANCHING LOGIC

Alternate Paths & Exception Loops (Q1 & Q2)

QUESTION 1: ACADEMIC EXCEPTION LOOPS

Committee Revision Loop (F5): When the committee demands revisions, an Exclusive Gateway routes to 'Send Revision Feedback'. The student receives the feedback via an Intermediate Message Catch Event, revises the proposal, and resubmits. To prevent infinite loops, the process enforces a maximum of 2 revision cycles before forcing an approve/reject decision.

Guide Decline & Re-Allocation Loop (F8/F9): When a faculty guide declines or the 3-day timer expires, the flow branches to Coordinator Manual Allocation. The coordinator can assign an alternate faculty member, appoint a co-guide, or place the team on a waitlist. If manual allocation succeeds, the flow rejoins confirmation; otherwise it ends at 'No Guide Available'.

QUESTION 2: BANKING EXCEPTION LOOPS

Underwriting & Collateral Loop (L7/L8): Borderline scores (600–749) divert from auto-approval to 'Underwriter Full Manual Review'. If unsecured credit is too risky, the underwriter requests collateral valuation. If the appraised Loan-to-Value (LTV) is insufficient, 'Re-Underwrite' loops back with adjusted loan amounts, preventing premature rejection.

Applicant Counter-Offer Negotiation (L10): Upon receiving the loan offer, the applicant can select 'Counter-Offer'. A Loan Officer manually reviews the requested rate/tenor adjustments. If accepted, negotiated terms are finalized; if rejected, the application concludes as rejected or withdrawn.

Disbursement Fault Retry (L12): An Interrupting Error Boundary Event catches RTGS/NEFT failures, routing to Operations for manual fallback retry without risking double debit.

FAULT ARCHITECTURE

Process Robustness, Recovery & Governance

🔁 RETRY LIMITS & LOOP CONTROL

Uncontrolled loops create infinite execution deadlocks. Both models enforce strict loop counters: Question 1 limits committee revision cycles to 2 iterations and document resubmissions to 2 attempts. Question 2 bounds credit bureau retries and applicant document corrections with explicit lapse timers.

⏰ SLA TIMERS & ESCALATION

Deadlines prevent applications from stalling silently. We use Interrupting Timers for hard cut-offs (Q1 14-day submission deadline, Q2 7-day offer acceptance expiry) and Non-Interrupting Timers for administrative reminders (Q1 10-day committee review SLA escalation to HoD, Q2 48-hour e-sign alert).

👥 HUMAN-IN-THE-LOOP FALLBACK

Automated services can yield false negatives. When automated DMN guide allocation in Q1 finds no available guide, it falls back to a Coordinator User Task. When automated KYC in Q2 fails, a Compliance Officer manually adjudicates the document, maintaining high conversion rates.

⚖️ FORWARD RECOVERY VS. TRANSACTIONAL COMPENSATION

In BPMN 2.0, Forward Recovery resolves active faults through retries and alternate branches (e.g., Q2 Core Banking disbursement error boundary). Backward Compensation cancels committed milestones. Q1 uses script compensation to release guide pre-allocations if a student withdraws.

MODEL GOVERNANCE

Design Decisions, Validation & Limitations

MODELLING RIGOR & VALIDATION PROOF

✓ **Strict Pool vs. Lane Boundary Discipline:** In BPMN 2.0, Sequence Flows can NEVER cross Pool boundaries. In Question 2, the external Credit Bureau is modelled as an independent Pool, and all communication uses Message Flows. In Question 1, internal lanes cleanly isolate academic roles.

✓ **Camunda Modeler 5.49 & bpmn-js Schema Validation:** Both `.bpmn` files were validated against official OMG BPMN 2.0 schemas. Diagrams parsed and rendered without XML syntax errors, containment flaws, or broken sequence references.

✓ **Deterministic Gateways:** Synchronized Parallel Joins prevent race conditions; Exclusive Gateways guarantee single-token determinism.

ACADEMIC LIMITATIONS & FUTURE PROPOSALS

🔍 **DMN Decision Table Expansion:** While Question 1 uses Business Rule Tasks for guide matching, Question 2 currently encapsulates risk scoring inside Service Tasks. Future work should extract banking scoring into external DMN decision tables.

🔍 **Global Event Sub-Processes for Cancellation:** Withdrawal is currently modelled at specific gateway points. Implementing an Interrupting Event Sub-Process with a Message Start Event would allow cancellation at ANY active stage in both systems.

🔍 **Transactional Sub-Processes for Disbursement:** Encapsulating disbursement, ledger reservation, and notification in a Transactional Sub-Process would provide atomic compensation guarantees.

PROJECT SYNTHESIS

Conclusion & Academic Defence

KEY ACADEMIC & PRACTICAL TAKEAWAYS

✓ **Comprehensive Dual-Domain Modelling:** Successfully modelled two diverse workflows—academic capstone allocation (Q1) and commercial retail lending (Q2)—demonstrating the flexibility of BPMN 2.0.

✓ **Complete Failure Path Coverage:** Engineered 11 failure paths for Question 1 and 12 failure paths for Question 2, covering data validation, timeouts, policy rejections, and system faults.

✓ **Advanced Execution Semantics:** Demonstrated mastery of Parallel Gateways (concurrency), Exclusive Gateways (decisions), Event-Based Gateways (competing events), and both Interrupting and Non-Interrupting Boundary Events.

✓ **Production-Ready Engineering:** Delivered validated BPMN 2.0 XML files, comprehensive Failure Path Registers, and full Task/Gateway Justification documents.

THANK YOU!

Questions & Academic Discussion

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